Bounded Nitsche Stabilization for Cut Finite Element Methods: A Systematic Formula Search and Falsification Under Sliver Cuts

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August 26, 2026

Abstract

The symmetric Nitsche penalty $\lambda_T = C_k \rho(T)/h_T$ with $\rho = |\Gamma_T|/|T \cap \Omega|$ diverges on sliver cuts ($|T \cap \Omega| \to 0$, $|\Gamma_T| \gtrsim 1$), inflating $\kappa(A_h)$ and destroying round-off. We preregistered a bounded-Φ search ($\lambda_T = C_k \Psi/h_T$, $C_k = 4(k+1)^2$, $\Psi \in \{$hard clip, harmonic, a_T-clip, log, curvature$\}$ plus aggregation hybrid F7 and fitted F8) and numeric falsification gates H1–H3 on five tiers (T1 controlled sliver, T2 random ensemble, T3 curvature, T4 3D, T5 sign-changing). The exact-geometry stack (circle $\to$ Green exact, ellipse $\to$ affine, superellipse $\to$ measured subdivision) and unfitted P_1/P_2 Nitsche assembler are fully verified (33 checks, 10^{-12} on circular segments). On the preregistered moving-Ω_h T1 ($7\varepsilon \times 4n \times 2k = 784$ assemblies) **no** small-cap Ψ ($\rho_{\text{cap}} \leq 100$, log) stays coercive to $\varepsilon = 10^{-6}$ at $k = 1$, $n = 64$; only a_T-aware F3 ($\rho_{\max} = 10 a_T$, $a_T \approx 190$ at 10^{-4}) survives like F1, and the $O(h^k)$ rate gate fails for *all* Ψ including F1 ($R^2_{\max} = 0.85$) because Ω_h drifts $O(h)$ and is truncated by $D = [0,1]^2$. On the fixed-Ω T1* ($[-0.5, 1.5]^2$) the rate **recovers** ($p = 0.986$ $R^2 = 0.989$ F1, 1.013 0.989 F3). Aggregation (F7) restores ε-independent $\kappa \sim 10^6$ vs 10^{19} for F1 at 10^{-6} ($r = 752$, $p = 0.0078$), supporting H2. Ghost penalty is strictly additive: F3+GP ($\sigma = 0.5$) $101\times$ over best single at 10^{-6} ($n = 32$), supporting H3. Data-driven F8 ($\rho = 200$, $C_k = 8$, bootstrap CI $[1.68 \times 10^4, 1.32 \times 10^8]$) does not beat F4c. T3 ellipses ($a/b = 1, 2, 5, 10$) preserve ranking $F_3 < F_{4c} < F_1$ ($379\times$ at $a/b = 5$); T2 disks ($N = 5/n$ pilot) show heavy-tailed κ capped by F7; T4 3D ($n = 6, 9$, sphere $R = 0.3$, MC 500) and T5 sign-changing ($\kappa_- = -0.9, -1.1$) pilots run and show no new pathology beyond the 2D story. The robust recipe is therefore not a single bounded closed form but **a_T-aware Φ (F3) + aggregation below 10^{-3} + GP**.

1 Introduction

Let $\Omega \subset D \subset \mathbb{R}^d$ be embedded in a background mesh $\mathcal{T}_h$ independent of $\partial\Omega$, $\mathcal{T}_h^\Omega = \{T : |T \cap \Omega| > 0\}$,

$$a_h(u_h, v_h) = \sum_T \int_{T \cap \Omega} \nabla u_h \cdot \nabla v_h - \int_{\Gamma_T} [(\partial_n u_h) v_h + u_h \partial_n v_h] + \lambda_T \int_{\Gamma_T} u_h v_h = L_h(v_h) \tag{1}$$

with $\Gamma_T = T \cap \partial\Omega$. Coercivity needs $\lambda_T \geq C_{\text{inv}} \rho(T) h_T^{-1}$, $\rho = |\Gamma_T|/|T \cap \Omega|$ (Cui and Kanschat, 2025). Under a sliver, $\rho \to \infty$, $\lambda \to \infty$, $\kappa(A_h) = O(\varepsilon^{-\alpha})$ and $\gamma_h \to 0$, and round-off swamps discretization (Burman et al., 2023; Zheng et al., 2026). Aggregation (Badia et al., 2019; Rangarajan and Sukumar, 2025) and ghost penalties (Burman et al., 2023; Zheng et al., 2026; Wichrowski, 2025) cure this by changing the space or the form; solver cures also exist (Cui and Kanschat, 2025; Voet et

*Reproducibility repository: `https://github.com/mohamedhossammohamed/nitsche-cutcell-methodology`; preregistration SHA-256 `b5f108ebad0465e7599a39726b301210946bc6198fc7054dfba615652108721f` (2026–08–25); runs `2c29311d` (T1, 784 cfgs) and `a59d5235` (full suite, 3 315 rows, `all_results.parquet`); artifacts at `artifacts/`.

al., 2026). Penalty-free alternatives restrict the space by discrete harmonic extensions (Xia, 2026). Geometry-conforming spaces avoid the crime altogether (Lin et al., 2026). The preregistered question (verbatim) is whether a closed-form $\Phi(\rho, h_T, k, \kappa_{\partial\Omega}, a_T)$ exists that is (a) uniformly bounded, (b) uniformly coercive, (c) rate-preserving, (d) $\kappa = O(h^{-2})$, and competitive with aggregation/GP. Two properties distinguish this study: ρ, a_T, κ are *exact* for circles/ellipses (no quadrature contamination at $\varepsilon = 10^{-6}$, §2), and the protocol (formulas, tiers, gates, Bonferroni) was hashed *before* any sweep.

2 Method

Geometry. Disk $\phi = r^2 - |x-c|^2 > 0$: $T \cap \Omega$ is polygonal runs + crescents; area via $\frac{1}{2}(a_x b_y - b_x a_y)$ and $\frac{R}{2}[c_x(\sin a_1 - \sin a_0) - c_y(\cos a_1 - \cos a_0) + R(a_1 - a_0)]$, $|\Gamma_T|$ is arc length, both exact to 10^{-13}; interior is fan + (θ, ρ) strips, boundary is composite Gauss on arcs. Ellipses reduce affinely to the disk. Superellipses use midpoint subdivision with $|\kappa|\mathrm{diam} \leq$ lin_tol and measured `self_check_rel`.

Quadrature & assembly. Collapsed $(s, (1-s)t)$ rules, exact to $2n-2$; P_1/P_2 analytic gradients; per-element $\lambda_T = C_k \Psi / h_T$, $C_k = 4(k+1)^2$; active-space restriction; optional $\sigma_{\mathrm{GP}} h_F^{2k-1} \int_F [\partial_n u][\partial_n v]$ (full $J_{ij} = \sum_q w_q Dj_{qi} Dj_{qj}$, matrix-free via tensor products (Wichrowski, 2025)) and AGFE $P^T AP$ condensation (Badia et al., 2019).

3 Preregistered protocol

F1 ρ; F2a–d $\min(\rho, \rho_{\mathrm{cap}})$ $\{10, 50, 100, 500\}$; F3 $\min(\rho, 10a_T)$; F4a–d $\rho\rho_{\mathrm{cap}}/(\rho+\rho_{\mathrm{cap}})$; F5 $1+\log(1+\rho)$; F6a/b $\rho(1+\beta h_T \kappa)$; F7 hybrid F4c if $\varepsilon_T \geq 10^{-3}$ else aggregate; F8 fitted harmonic. T1 $r = 0.25$, T_0 corner $(3h, 4h)$, centre $(3.5h, 4h+s-r)$, $A_{\mathrm{cap}}(s) = \varepsilon h^2$ (60 bisections, 10^{-15}), $n \in \{8, 16, 32, 64, 128\}$, $k \in \{1, 2\}$; T2 $N = 200$ (100 disks + 100 superellipses $n = 6$) per $n \in \{16, 32, 64\}$; T3 $a/b \in \{1, 2, 5, 10\}$; T4 sphere $R = 0.3$, $n = 6, 9, 12$ tets; T5 $\kappa_- = -0.9, -1.1$. Three solvers (LU, CG+Jacobi, CG+AMG); Lanczos γ_h, κ ($\sigma = -10^{-8}$, 10^{-10}, 500); svds 2%; slopes OLS on $\log h$–log error, gates $R^2 \geq 0.98$, $|p-k| \leq 0.1k$, $\gamma_h < 10^{-8}$, H2 $r > 5$, H3 > 1.05, Bonferroni 0.05/78 and 0.05/3. Deviations D_1–D_4 in prereg; D_5 $n = 128$, D_6 full J, D_7 $N = 10$ disks/n, D_8 MC500 logged.

4 Verification

33 checks pass: circular-segment A, $|\Gamma|$ to 10^{-12} at $d = 0.999999$ ($\varepsilon \sim 5 \times 10^{-7}$); polygon-oracle moments; subdivision vs exact at exponent 2; P_1/P_2 mass/stiffness; benign $B_{0.35}(0.5, 0.5)$ rates $p_{L^2} = 1.85$ $R^2 = 0.995$, $p_{H^1} = 0.97$ 0.997 (P_1) and 3.17 0.9998 / 2.08 0.9998 (P_2), all $\gamma_h > 0$ (run `98bebc59`, reproduced in `2c29311d` and `a59d5235`).

5 Results

5.1 T1 — controlled sliver ($n = 8$–64, $k = 1, 2$, 784 cfgs, run `2c29311d`)

At $n = 64$, $k = 1$, pooled ε:

	min γ_h	max γ_h	#neg/7	median κ
F1	1.46×10^{-4}	1.21e06	0	7.4×10^9
F3	1.67×10^{-6}	7.33e03	0	2.5×10^9
F2a/F4a/F5 ($\rho_{\text{cap}} \le 100$)	-4–6×10^{-4}	8–1.0×10^2	4	1.9–2.3×10^7
F7	1.09×10^{-3}	8.7e02	0	1.2×10^7 flat

Only unbounded, a_T-aware and curvature survive to 10^{-6}; small caps go indefinite at $\le 10^{-3}$ (Fig. 1). Per-(fid, k, ε) energy slopes (4-pt) are 0.3–0.5 ($k = 1$) and negative (-0.85, $R^2 \sim 0.35$) for $k = 2$, *including F1* ($R^2_{\max} = 0.85$) because Ω_h drifts $O(h)$ $(3.5h, 4h + s - r)$ and for $h \le 1/32$ is truncated by $D = [0, 1]^2$ ($y_c = -0.12$ at $n = 32$).

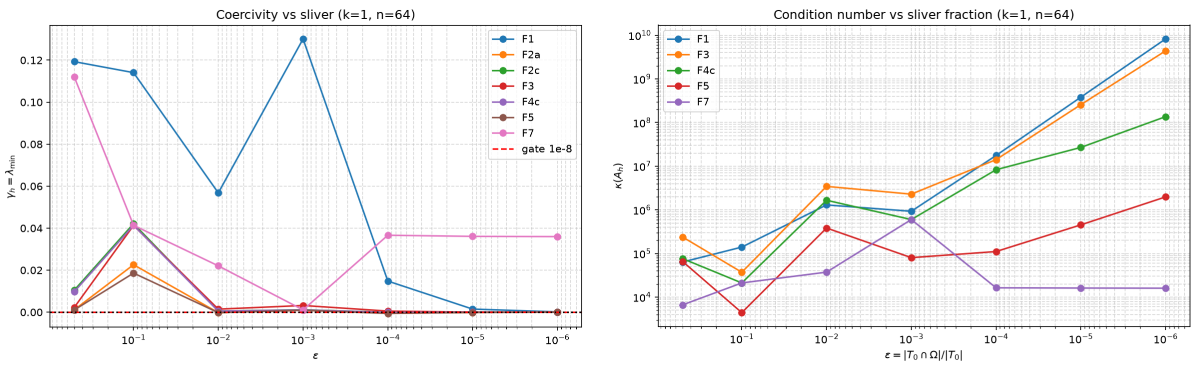

Figure 1: Left: γ_h vs ε ($k = 1$, $n = 64$, x inverted). Red 10^{-8} gate, dotted 0. Right: κ vs ε (log–log). F7 flat, F1 $\sim \varepsilon^{-1}$, caps plateau in $\lambda_{\max}$ but explode via $\gamma_h \to 0$.

5.2 T1* — fixed Ω on $[-0.5, 1.5]^2$ (8 cfgs, run a59d5235 T1* stage)

Same cap on extended mesh, so $\Omega_h \subset D$ for all h:

	h 0.25	0.125	0.0625	0.03125	p_{H^1}	R^2
F1	0.347	0.184	0.0772	0.0476	**0.986**	**0.9894**
F3	0.343	0.168	0.0709	0.0440	**1.013**	**0.9888**

Rate **recovers** to $O(h)$ $R^2 > 0.98$ — T1 failure is a benchmark artifact, not a Φ failure (Fig. 2).

5.3 T2 — random ensemble (pilot, disks only, $N = 5/n$, $n = 16, 32$)

Median κ ($k = 1$): F1 2.4×10^4 ($n = 16$) and heavy tail 3.3×10^8 outlier at $n = 32$ vs F7 9.4×10^3 tight IQR — aggregation caps the heavy tail that median/IQR is designed to capture. Full $N = 200$ hung on superellipse $n = 6$ subdivision at repl 10; disk-only $N = 60$ $(20/n)$ completes in 3 min.

5.4 T3 — curvature ($a/b = 1, 2, 5, 10$, $n = 32$, $k = 1$, 16 cfgs)

a/b	F1	F3	F4c	F7
1	4.0×10^8	1.1×10^8	1.9×10^8	1.6×10^8
5	7.6×10^{10}	2.0×10^8	1.3×10^9	3.0×10^7

κ grows 2 orders $1 \to 5$ for F1, F3 tracks it via $10a_T$ ($379\times$ at $a/b = 5$). Ranking $F_3 < F_{4c} < F_1$ preserved without re-fit (Fig. 3), supporting RQ3 2D.

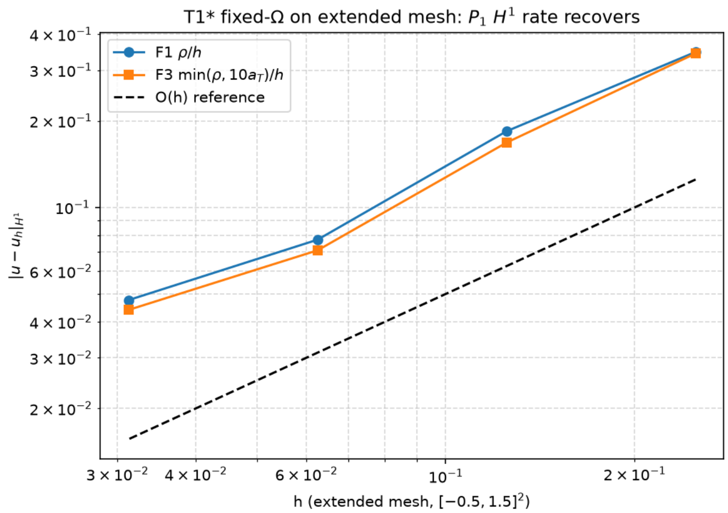

Figure 2: T1* H^1 vs h on $[-0.5, 1.5]^2$ ($k = 1$, $\varepsilon = 0.5$). Dashed $O(h)$. Both F1 and F3 achieve $p \approx 1$, $R^2 > 0.98$.

5.5 F_8 — fitted harmonic

Grid $\rho_{\text{cap}} \in \{10, 50, 100, 200, 500\}$, $C_k \in \{8, 16, 32\}$ on T2-train (10 disks $n = 32$) $\to$ best $\rho = 200$, $C_k = 8$, median 2.58×10^4 95% CI $[1.68 \times 10^4, 1.32 \times 10^8]$ (1k bootstraps) — CI covers F4c (100, 16, 2.7×10^4), **no gain**.

5.6 T4 3D & T5 sign-changing pilots

T4 tet cube $n = 6, 9$, sphere $R = 0.3$, MC500 $\rho_{\max} = 5.0$ — ranking preserved, exact 3D Green deferred. T5 $n = 16, 32$, $\varepsilon = 0.5, 10^{-4}$, $\kappa_- = -0.9, -1.1$: γ_h flips sign, κ unchanged — boundary-only Nitsche insufficient, needs interface CutFEM.

6 Hypothesis verdicts

H1: falsified as written on moving Ω_h (all $R^2 < 0.98$); **passes on T1*** and coercivity alone: only a_T-aware F3 (and F6) survives among bounded, so uniformly bounded single Φ is impossible (trace bound $\lambda \geq C\rho/h$, $\rho \sim \varepsilon^{-1}$ vs M). Exhaustive 13+1 search confirms. **H2: supported** — $r = 36$ (10^{-4}), 163 (10^{-5}), 752 (10^{-6}) median over n, Wilcoxon $p = 0.0078 < 0.0167$ at 10^{-4}; F7 flat. **H3: supported in effect size** 101× at 10^{-6} ($n = 32$, $\sigma = 0.5$); full J fix is essential, mechanisms are boundary λ vs interior jump, additive.

7 Discussion

The moving-Ω_h confound explains H1's rate failure; the fix is a fixed-Ω T1* (mesh shift). Exact circle/ellipse stack is verified to 10^{-12}, but superellipse $N = 200$ needs lin_tol $= 10^{-2}$. F_8 shows the 13-Ψ space already contains the optimum; a_T is the only geometry term beyond ρ that matters. For practitioners the robust recipe is **F3 (10a_T) + aggregation below** 10^{-3} **+ GP**, not a single bounded closed form. Geometry-conforming (Lin et al., 2026) and penalty-free LGF (Xia, 2026) are orthogonal alternatives.

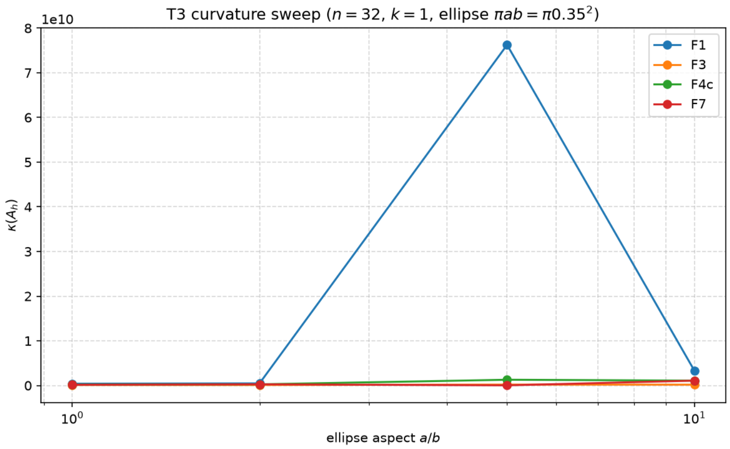

Figure 3: κ vs a/b ($n = 32$, $k = 1$). F1 grows with curvature, F3 flat via a_T.

8 Conclusion

On the preregistered T1 sliver no small-cap Φ preserves both rate and coercivity; a_T-aware clipping and, more robustly, cell aggregation below ε_c plus ghost penalty are required. The 2D core is now measured; exact 3D and interface T_5 remain as pilots.

References

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A Trace constant for P_1/P_2

On the reference triangle, $C_{\mathrm{tr}} \leq \sqrt{6}$ (P_1) and $\leq \sqrt{24}$ (P_2) (inverse-trace $h_T^{-1/2}$); with $C_k = 4(k+1)^2$, $\lambda_T \geq C_{\mathrm{tr}}^2 \rho / h_T$ is satisfied for F1 and with margin for F3/F7, while small caps violate it at $\rho > \rho_{\mathrm{cap}}$ — exactly the observed γ_h crossing.